

Maike L Morrison

maike@santafe.edu | MaikeMorrison.com

Appointments

2025– **Santa Fe Institute** **Santa Fe, NM**
Omidyar Postdoctoral Research Fellow

Education

2020–2025 **Stanford University** **Stanford, CA**
PhD in Biology (Ecology & Evolution)
Dr. Noah Rosenberg, advisor
Associate Teaching Certificate

2016–2020 **The University of Texas at Austin** **Austin, TX**
BS in Mathematics, Dean's Scholars Honors Program
Drs. Lauren Ancel Meyers & Mark Kirkpatrick, advisors

Research Experience

Mar 2023– **Visiting Scientist, International Agency for Research on Cancer** **Lyon, France**
Sep 2023 *Dr. Nicolas Alcalá, International Agency for Research on Cancer*

Jun 2022– **Complex Systems Summer School Student** **Santa Fe, NM**
July 2022 *Santa Fe Institute*

May 2019– **UCSF Amgen Scholars Program** **San Francisco, CA**
Jul 2019 *Dr. John Witte, UC San Francisco Department of Biostatistics & Epidemiology*

Jun 2018– **The Mathematical Biosciences Institute NSF REU** **State College, PA**
Aug 2018 *Dr. Ephraim Hanks, Penn State University Department of Statistics*

Peer-Reviewed Publications

6. **Morrison ML**, Xue KS, Rosenberg NA (2025). Quantifying compositional variability in microbial communities with FAVA. *Proceedings of the National Academy of Sciences*, 122(11), e2413211122.
5. **Morrison ML** & Rosenberg NA (2023). Mathematical bounds on Shannon entropy given the abundance of the *i*th most abundant taxon. *Journal of Mathematical Biology*, 87(5), 76.
4. **Morrison ML**, Alcalá N, Rosenberg NA (2022). FSTRUCT: an F_{ST} -based tool for measuring ancestry variation in inference of population structure. *Molecular Ecology Resources*, 22(7), 2614-2626.
3. Ingle T, **Morrison ML**, Wang X, Mercer T, Karman V, Fox S, Meyers LA (2021). Projecting COVID-19 isolation bed requirements for people experiencing homelessness. *PLOS One*, 16(5), e0251153.
2. Kachuri L*, Francis SS*, **Morrison ML**, Wendt GA, Bossé Y, Cavazos TB, Rashkin SR, Ziv E, Witte JS (2020). The landscape of host genetic factors involved in immune response to common viral infections. *Genome Medicine* 12(93), 1-18.
1. **Morrison M***, Castro LA*, Meyers LA (2020). Conscientious vaccination exemptions in kindergarten to eighth-grade children across Texas schools from 2012 to 2018: A regression analysis. *PLOS Medicine* 17(3), e1003049.

Preprints

4. Sexton-Oates A*, Mathian E*, Candeli N, ..., **Morrison ML**, ..., Alcalá N, Foll M, Fernandez-Cuesta L (2025) A clinically relevant morpho-molecular classification of lung neuroendocrine tumours. *medRxiv*.

3. **Morrison ML**, Mangé L, Senkin S, Rosenberg NA, Foll M, Fernandez-Cuesta L, Alcala N (2023). Variability of mutational signatures is a footprint of carcinogens. *medRxiv*.
2. Xue KS, Walton SJ, Goldman DA, **Morrison ML**, Verster AJ, Parrott AB, Yu FB, Neff NF, Rosenberg NA, Ross BD, Petrov DA, Huang KC, Good BH, Relman DA (2023). Prolonged delays in human microbiota transmission after a controlled antibiotic perturbation. *bioRxiv*.
1. Miyagi M, **Morrison M**, Kirkpatrick M (2022). How many ecological niches are defined by the super-abundant marine microbe *Prochlorococcus*? *bioRxiv*.

*These authors contributed equally

R Packages

FAVA: An F_{ST} -based Assessment of Variability over relative Abundances

sigvar: Quantify and visualize variability of mutational signatures within and across samples

FSTruct: Measure variability in population structure estimates

Teaching

Winter 2025	BIO 192/387: Statistical Methods in Ecology & Evolution <i>Instructor of Record</i>	Stanford University <i>Faculty advisor: Dr. Noah Rosenberg</i>
	· Designed and taught course on the theory and application of biological diversity statistics.	
Fall 2023	BIO 388: CEHG Seminar - Microbiome Ecology & Evolution <i>Instructor of Record</i>	Stanford Univ. <i>Co-taught with Dr. Noah Rosenberg</i>
	· Co-organized Fall 2023 Computational, Evolutionary, & Human Genomics seminar.	
	· Facilitated accompanying class, including lecturing on statistical methods for biology.	
Winter 2022	BIO 187: Mathematical Population Biology <i>Graduate Teaching Assistant</i>	Stanford University <i>for Dr. Noah Rosenberg</i>
	· Prepared and presented lectures on mathematical bounds on F_{ST} and Shannon entropy.	
Winter 2021	BIO 85: Evolution <i>Graduate Teaching Assistant</i>	Stanford University <i>for Drs. Dmitri Petrov & Molly Schumer</i>
	· Wrote & graded assessments and organized discussion sections for introductory online course.	

Awards

2025	Samuel Karlin Thesis Prize in Mathematical Biology
2024-2025	Stanford University Centennial Teaching Assistant Award
2022	Stanford Biology Department Excellence in Teaching Award
2020	Dean's Honored Graduate: University of Texas College of Natural Sciences
2020	George H. Mitchell Award for Outstanding Undergraduate Research
2019-2020	Alan Kaylor Cline Dean's Scholars Scholarship
2019-2020	Phi Beta Kappa, junior year election
2019	Barry M. Goldwater Scholarship

Research Fellowships

2025–2028	Santa Fe Institute Omidyar Postdoctoral Research Fellowship
2021–2025	NSF Graduate Research Fellowship Program
2020–2025	Stanford Graduate Fellowship

Academic Service

2024–2025	Member, CEHG Symposium Organizing Committee	Stanford University
	<i>Center for Computational, Evolutionary, and Human Genomics (CEHG)</i>	
2023–2024	Student Member, Ecology & Evolution PhD Admissions Committee	Stanford University
2021–2022	Co-Organizer, Eco-Evo Lunch	Stanford University
2021	Co-Organizer, Biology Pre-Orientation Program (BioPOP)	Stanford University

Peer review activity: *Genetics, Molecular Biology & Evolution, Theoretical Population Biology*

Mentorship: One undergraduate researcher, two undergraduates through the Stanford Women's Community Center, several prospective/incoming graduate students through the Biology Preview and ADVANCE programs.

Conference Presentations**Selected Oral Presentations**

May 2024	Modeling & Theory in Pop. Bio. (Banff Int. Research Station)	Banff, AB
	“Quantifying the stability of microbiomes and the timescale of antibiotic perturbation”	
	<i>Award for best talk by graduate student or postdoc</i>	
May 2024	American Mathematical Society Spring West. Sect. Meeting	San Francisco, CA
	“Quantifying compositional variability in microbial communities”	
	<i>Invited talk</i>	
Aug 2023	Ecological Society of America Annual Meeting	Portland, OR
	“Quantifying compositional variability in microbial communities”	

Selected Poster Presentations

June 2025	Society for Modeling & Theory in Population Biology	Chicago, IL
	“A general F_{ST} framework reveals the variability of rare versus common alleles”	
	<i>Award for Outstanding Poster</i>	
Dec 2022	The Human Microbiome: Ecology and Evolution (Keystone Symposia)	Banff, AB
	“A diversity statistic for quantifying microbiome stability”	
June 2022	Population, Evolutionary, & Quantitative Genetics Conference	Pacific Grove, CA
	“FSTruct: An F_{ST} -based tool for quantifying ancestry variability”	
	<i>Genetics Society of America Poster Award</i>	
June 2019	Evolution Meeting	Providence, RI
	“How many ecotypes are comprised by the world's most abundant photosynthetic organism?”	
	<i>Undergraduate Diversity at Evolution Award Program</i>	